IRRIGATION WATER MANAGEMENT · NOTE IM02

Surface Irrigation II — Furrow, Contour-Furrow, Corrugation and Sub-Irrigation

Course topics: IDE5.5 L16, CAE1.3 L12, IDE-501 T9

Furrow irrigation

Used for row crops — maize, sorghum, sugarcane, cotton, tobacco, groundnut, potatoes and vegetables — with furrows formed between the rows during planting and cultivation. Usually one furrow per crop row; one for two rows in some vegetables; two or more per row in orchards and widely spaced vegetables. Size and shape follow the crop, the equipment and the row spacing.

- **Application:** run small streams down the furrows; water soaks in and spreads sideways to wet the ridges. The run time depends on the water needed to refill the root zone, the infiltration rate and the lateral spread. Large or small supplies both work — vary the number of furrows irrigated at once. Furrows also drain rainfall quickly where surface drainage is needed.
- **Shape:** wide, shallow furrows on slowly absorbing soils (the general preference); narrow, deep furrows in long runs on permeable soils to limit percolation at the upper end.

Advantages

- Water touches only one-fifth to one-half of the surface, so less puddling, crusting and evaporation; earlier cultivation possible (important on heavy soils).
- Fits a wide range of slopes without erosion; especially good for crops injured by contact with water (e.g. maize).
- Less labour for land preparation and irrigation; no land lost to field channels (unlike check basins); large or small volumes usable by dividing the flow among furrows.
- **Grading:** grade the land so water travels the full furrow length without ponding — remove highs and lows and give enough slope; careful tillage then keeps the grade. Furrows usually run down the slope, or on the contour to control erosion.
- **Hydraulics:** unsteady open-channel flow, decreasing downstream as water infiltrates; the wetting pattern differs from borders and basins because water also spreads sideways. Dominant variables: entrance stream size, infiltration rate, size and shape of the wetted section, furrow slope and hydraulic resistance (surface roughness and crop).
- **Infiltration opportunity:** the time of ponding (difference between advance and recession times) should equal the time to infiltrate the net depth and bring the root zone to field capacity, and be uniform along the furrow.

Design

Select furrow spacing, length, slope, stream size and duration to suit the soil and topography.

Factor	Recommendation
Spacing	Potatoes, maize, cotton: a furrow between each row; vegetables (lettuce, carrot, onion): two or more rows between furrows; orchards: furrows 1 to 2 m apart
Depth and size	Vegetables 7–12 cm deep; orchards and some row crops 20–30 cm; common row-crop furrow about 25 cm wide and 8 cm deep. Wide shallow furrows for slowly absorbing soils; narrow deep ones for long runs in permeable soils
Grade	Minimum 0.05% (for surface drainage); generally 0.2–0.5%. Steeper grades lower infiltration and lateral spread, waste water at the end and raise velocity and erosion — use contour furrows on steep land. Keep the slope uniform for uniform wetting
Stream size	Usually 0.5 to 2.5 L/s, larger on lighter soils. Start with the largest non-erosive stream; once water reaches the end, reduce it (**cut-back stream**) to keep wetting with minimum waste
Length of run	The longest furrow irrigable safely and efficiently: about 45 m where infiltration exceeds 5 cm/hr, up to 300 m or more on heavy, slowly permeable soils. Too long → over-irrigation at the top and under-irrigation at the bottom; too short → more cross ditches and labour

- **When to make furrows:** before planting, at planting or after the plants are up — often small furrows for young plants, enlarged later; always big enough for the stream.
- **Crop position:** usually on top of the ridge; in deep sands, seedlings go near the bottom of small furrows — or transplant vegetables in the furrow, irrigate once or twice, then form furrows between the rows.
- **Tools:** double mould-board plough, lister, wooden plough with a furrower, and the disc-drum corrugator-furrower for small furrows in sandy soils.
- **Black cotton soils:** combine borders with furrows — these soils crack badly, and the furrows stop water escaping between furrows, drain heavy rain evenly and limit erosion; the border ridges must be big enough to confine the flow.

Contour furrows

- Water flows across the slope in furrows with a gentle uniform grade, fed from supply channels running across the slope or downhill with drop structures.
- Usable on nearly all irrigable soils — slopes up to 5% on light soils and up to 20% in stable orchard soils with conservation measures; not on steep heavy black soils unless the furrows lie within contour border strips.
- **Layout:** prepare a contour map, locate the supply channels on non-erosive or controlled grades, lay drains that do not hinder farming, fix the furrow and planting lines, and smooth the field with grading and a float. Row-crop contour furrows are made every year; orchard furrows and channels are permanent.
- **Survey:** take a staff reading at the start and mark points every 15 m, giving the right gradient between readings.

Corrugation irrigation

- Corrugations are small furrows for close-growing crops — hay, pasture and small grains — running down or across the slope; crop rows need not match them.
- **Spacing** depends on the soil — closer on more porous soils: typically 45 to 90 cm, 35 to 40 cm on impervious soils or sloping land. In sandy soils water goes deep with little sideways spread, so keep them close.

- **Preparation:** prepare the soil and sow without regard to the irrigation layout, then make the corrugations with a corrugator after sowing but before germination; control the flow as in furrows.

Sub-irrigation

- Water is applied below the surface by maintaining an artificial water table; it reaches the roots by capillarity. It enters through deep trenches or underground pipes (tile or mole drains): trenches 30 cm to 1 m deep, 15–30 m apart, with field supply channels, ditches or trenches and drainage channels for excess water.
- **Needs** complete control of the water table; uniform soil, permeable horizontally and vertically; a barrier to deep percolation (an impermeable layer or a naturally high water table); and smooth, nearly level land.
- **Suits** soils of low water-holding capacity and high infiltration; evaporation loss is minimal. Saline water cannot be used.
- **In India** its use is limited — vegetables near Dal Lake, Kashmir, and coconut palms in Kuttanad, Kerala (irrigated through open ditches).

Sources: ICAR eCourse 'Irrigation Engineering' (IIT Kharagpur) — surface, sprinkler and drip irrigation lessons; ICAR eCourse 'Micro Irrigation Systems Design' (AgriMoon). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. In furrow irrigation, water contacts about ___ of the land surface

(a) 1/10 to 1/5 (b) 1/5 to 1/2
(c) 3/4 (d) all

Q2. Generally satisfactory furrow grade is

(a) 0.01–0.05% (b) 0.2–0.5%
(c) 1–2% (d) 5%

Q3. Usual furrow stream size is

(a) 0.05–0.25 L/s (b) 0.5–2.5 L/s
(c) 5–25 L/s (d) 50 L/s

Q4. Reducing the furrow stream after water reaches the end is called

(a) surge (b) cut-back
(c) ponding (d) check

Q5. Typical spacing of corrugations is

(a) 10–15 cm (b) 45–90 cm
(c) 2–3 m (d) 10 m

Q6. Sub-irrigation in India is practised for coconut palms in

(a) Konkan (b) Kuttanad
(c) Kutch (d) Kashmir

Answers

Q1 — (b) 1/5 to 1/2
Q2 — (b) 0.2–0.5%
Q3 — (b) 0.5–2.5 L/s
Q4 — (b) cut-back
Q5 — (b) 45–90 cm

Q6 — (b) Kuttanad